

LangChain & Agentic AI – From Basics to Advanced

demo- 3 Notes

Without LangChain:

We can develop AI based applications.

But we (Our Application) are responsible for

Application

1. Prompt Management
2. Model Management
3. Conversation History and State Management
4. Handling our own private data
5. Tool Management like Calculator, Search, Database
6. Agentic Activities

With LangChain:

Application

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LangChain

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1. Prompt Management
2. Model Management
3. Conversation History and State Management
4. Handling our own private data
5. Tool Management like Calculator, Search, Database
6. Agentic Activities

LangChain provides useful building blocks, abstractions and Integrations for creating LLM applications.

```
import os # os is module
```

```
from langchain_openai import ChatOpenAI
```

```
OPENAI_API_KEY = os.getenv("OPENAI_API_KEY") #function
```

```
llm = ChatOpenAI(model="gpt-4o",api_key=OPENAI_API_KEY)
```

```
question = input("Enter the question: ")
```

```
response = llm.invoke(question)
```

```
print(response.content)
```

```
import os
```

```
from langchain_google_genai import ChatGoogleGenerativeAI
```

```
GOOGLE_API_KEY = os.getenv("GOOGLE_API_KEY")
```

```
llm = ChatGoogleGenerativeAI(
```

```
model="gemini-3.6-flash",  
google_api_key=GOOGLE_API_KEY  
)
```

```
question = input("Enter the question: ")
```

```
response = llm.invoke(question)
```

```
#print(response.content)
```

```
print(response.content[0]["text"])
```